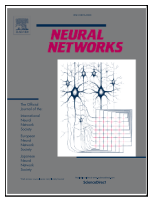
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Full Length Article



Memory dendritic neural P systems for multi-ground type fault line selection in distribution networks

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ABSTRACT

Rapid and accurate single-line-to-ground (SLG) fault line selection in distribution networks is crucial to power system safety. Although existing deep learning methods have achieved good performance, there is still room for improvement in the accuracy of fault line selection when faced with multiple neutral grounding types and varying fault conditions. The neural-like P systems use the "point neurons" computing mode and fails to fully utilize the multi-dendritic branching characteristics of biological neurons, resulting in limited computing efficiency. Therefore, this paper proposes a novel memory dendritic neural P (MDNP) system, which effectively improves the system's fault feature learning ability under different grounding types by introducing the triggering rules of memory dendritic factors, dynamically adjustable dendritic structure and adaptive learning mechanism, significantly reduces computing resource consumption and accelerates convergence speed. Furthermore, the design of online application scheme enables the MDNP system to achieve online optimization of fault line selection in dynamic power grid. The experimental results show that compared to seven baseline models, the MDNP system achieves fault line selection accuracy of 98.2% and 97.3% under non-grounding and arc suppression coil grounding types, respectively. The average fault line selection time is only 1.78s. Additionally, the computational resource consumption is reduced by 56.6% compared to traditional neural-like P systems, demonstrating its exceptional performance and potential for fault line selection tasks.

1. Introduction

Nowadays, distribution networks are increasingly confronted with various risk challenges, among which addressing single-line-to-ground (SLG) faults constitutes one of the critical issues (Deng et al., 2025). In China, distribution networks with voltage levels of 3 to 35 kV typically use either a neutral non-grounding system or an arc suppression coil grounding system (Yuan et al., 2023). SLG faults are the most common, accounting for over 80% of all distribution network faults, which pose serious risks to power system safety and the safety of local inhabitants (Hao et al., 2022; Yuan et al., 2025). Therefore, it is essential for power personnel to quickly and accurately identify lines with SLG faults (Guo et al., 2025a).

Fault line selection involves identifying the faulty line by monitoring characteristic differences in the distribution network during an SLG fault (Gao et al., 2025). Currently, traditional methods detect faults by analyzing steady-state and transient zero-sequence current or voltage differences between lines (Hong et al., 2024; Wang et al., 2025; Zhang et al., 2025). These methods require manually extracting fault characteristics and constructing fault line selection criteria (Zhang et al., 2024). However, these methods require dedicated zero-sequence signal measurement devices, which increases equipment costs. They also rely on expert-defined fault detection criteria, which may produce inconsistent identification results (Guo & Shi, 2023).

In recent years, machine learning has been widely applied to fault line selection in distribution networks (Deng et al., 2021; Gao et al., 2021; Guo et al., 2025a). Existing research has primarily focused on machine learning methods based on zero-sequence signal analysis. For instance, Deng et al. (2021) proposed a zero-sequence voltage analysis method that combines a clustering algorithm with long short-term memory (LSTM) networks. Gao et al. (2021) improved system fault tolerance by fusing zero-sequence current waveforms and utilizing convolutional neural networks (CNNs) for feature extraction. Guo et al. (2025a) introduced a Swin-Transformer model with Fourier

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preprocessing to achieve threshold-free fault line selection. However, these methods still share similarities with traditional fault line selection techniques and rely on the installation of specialized zero-sequence signal measurement devices, which increases system complexity and costs. Furthermore, the performance of existing methods varies significantly across different neutral grounding systems (Liang et al., 2023). In particular, in systems grounded by arc suppression coils, the partial compensation of capacitive currents can attenuate fault characteristics, severely affecting the accuracy of fault line selection (Lin et al., 2024). Additionally, current machine learning-based fault line selection methods primarily rely on offline recorded data for model training and testing, failing to fully leverage real-time operation data for online model updates. This limitation hinders the adaptive capability of these methods. For example, Zhang et al. (2021) proposed a scheme combining generative adversarial networks (GANs) and deep separable convolutional networks (DSCNs), while Xiao et al. (2024) developed a modified convolutional neural networks (MCNNs), but neither considered an online learning mechanism using real-time data. In summary, current fault line selection technologies in distribution networks face three key challenges: 1) Zero-sequence signal-based methods require additional specialized measurement devices, increasing system complexity and costs; 2) Existing machine learning methods need to improve fault line selection accuracy, particularly in different grounding systems (especially arc suppression coil grounding systems); and 3) Current models lack mechanisms for online learning with real-time data, limiting their dynamic adaptability. To address these issues, this paper aims to develop a fault line selection method for distribution networks that does not rely on zero-sequence signals and is adaptable to various grounding systems, thereby improving both fault line selection accuracy and adaptability.

The Spiking Neural P (SNP) system is regarded as the third-generation neural network model, inspired by the firing mechanism of biological neurons (Zhao & Liu, 2024). By making full use of the characteristics of large-scale parallel computing, distributed information processing, and good scalability, the SNP system significantly improves the performance of complex computing tasks. The theoretical research on the SNP system mainly focuses on three aspects: creating new membrane objects, constructing new membrane structures, and designing new membrane rules. The aim is to enhance the computational capability, flexibility, and applicability of membrane systems (Peng et al., 2017; Yin et al., 2024). In terms of applications, the SNP system has been successfully used in areas such as power system fault diagnosis (Hu et al., 2024; Ma et al., 2024; Wu & Wang, 2025), time series analysis (He et al., 2024), image processing (Li et al., 2025; Xia et al., 2025), and engineering optimization (Liu et al., 2024; Zhu et al., 2021). Yu et al. (2025) utilizes a self-synaptic SNP system to achieve power balance and coordinated control of multiple microgrids. Hu et al. (2024) proposes an intelligent fault line selection method based on a learning-type SNP system, effectively improving fault detection accuracy. However, existing SNP systems generally adopt simplified point neuron models without in-depth research on the computational capabilities of subcellular compartments (such as neuronal dendrites) (Gidon et al., 2020). In fact, biological neurons have highly differentiated dendritic structures, containing thousands of functionally independent subcellular compartments (such as dendritic spines and dendritic branches) that can independently perform nonlinear operations (Pagkalos et al., 2023). Neuroscience research has shown that dendritic branching forms a natural hierarchy for information processing: distal dendrites selectively process specific input features, whereas proximal dendrites integrate higher-level signals (Xu et al., 2025). Therefore, this paper aims to develop a neural P system based on dendritic computing to accurately fault line selection under different grounding types.

In previous work, the proposed integrated dynamic SNP (IDSNP) system (Ma et al., 2024) demonstrated the capability to perform distribution network fault line selection through neuron-level computation. However, this system still exhibits limitations in computational resource optimization, particularly in terms of requiring further improvements in neuronal configuration efficiency. Meanwhile, the system has poor adaptability to different grounding types, resulting in the accuracy of fault line selection can be further improved. In addition, due to the lack of a dynamic optimization mechanism based on real-time data, the system shows obvious application limitations when dealing with dynamically changing power grid environments.

To address the above challenges, this paper proposes a memory dendritic neural P (MDNP) system, which effectively breaks through the limitations of traditional SNP systems that are limited to point neuron processing by simulating the hierarchical computing characteristics of biological neuron dendrites. The system achieves storage and integration of historical information through designed memory dendrite factors, thus enhancing the fault feature learning capability. In particular, the dendrite multi-branch computing structure realizes the processing of different grounding types (including non-grounding and arc suppression coil grounding) through a dynamic adjustment mechanism, which significantly improves the accuracy of fault line selection for multi-ground type. On this basis, the grounding type is obtained based on the common information model (CIM) and a simulation model is established to achieve data augmentation. Furthermore, an adaptive learning mechanism is used to improve the convergence speed and detection accuracy of the MDNP system. Finally, through the developed online application scheme, the system has been successfully applied in engineering, verifying its effectiveness in a dynamic power grid environment.

The main contributions of this paper are as follows:

(1) This paper proposes a novel MDNP system that is an improvement over the traditional SNP system. Unlike SNP systems restricted to parallel processing of point neurons, the MDNP system incorporates an innovative dendritic hierarchical computing architecture that enables sophisticated parallel computation capabilities.
(2) A multi-ground type fault line selection method is developed based on the MDNP system. The memory dendritic factor trigger rule and dynamic dendritic structure are introduced to store and integrate historical information, enhance the fault feature learning and adaptive adjustment capabilities, and significantly improve the fault line selection accuracy under different neutral grounding types.
(3) The paper establishes a practical online application scheme of MDNP system for fault line selection, enabling real-time optimization and direct engineering application in distribution networks. Furthermore, an adaptive learning mechanism is introduced when the MDNP system fault line selection, which effectively improves the convergence speed.

The structure of the rest of this paper is as follows: Section 2 reviews the fault line selection problem and the numerical SNP system; Section 3 presents the designed the MDNP system; Section 4 introduces the application of the MDNP system to fault line selection in distribution networks; Section 5 provides the experimental validation of the proposed method. Finally, the conclusions of this paper are presented.

2. Preliminaries

This section briefly reviews the fault line selection problem under different grounding types and defines the numerical SNP system.

2.1. SLG fault analysis of neutral non-grounding and arc suppression coil grounding systems

The SLG fault analysis in the neutral non-grounding and arc suppression coil grounding systems is shown in Fig. 1 (Gao et al., 2021). When K is open, it represents a neutral non-grounding system. The voltage and current vector diagram of an SLG fault on Line 2 is shown in Fig. 2. At this point, the phase-to-ground voltages are given as: $\dot{U}_A = 0, \dot{U}_B = \sqrt{3}\dot{E}_A e^{j\frac{-5\pi}{6}}, \dot{U}_C = \sqrt{3}\dot{E}_A e^{j\frac{5\pi}{6}}$. The zero-sequence voltage and current of phase A are given by Eqs. (1) and (2), respectively. In

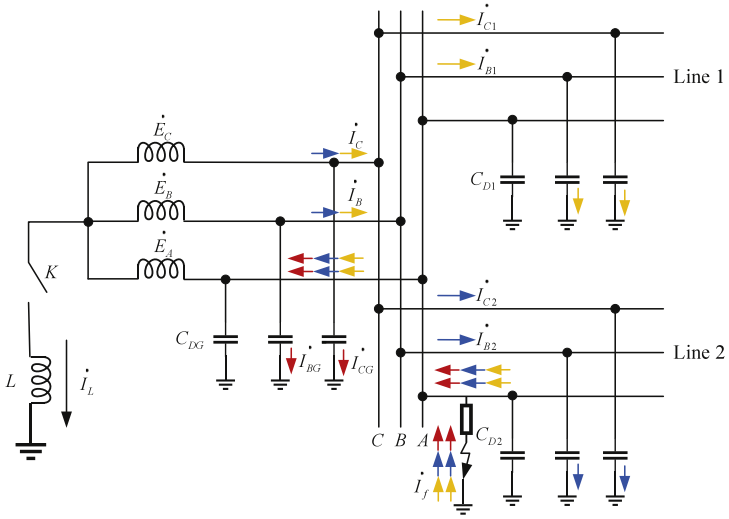


Fig. 1. SLG fault occurs in the neutral non-grounding and arc suppression coil grounding systems.

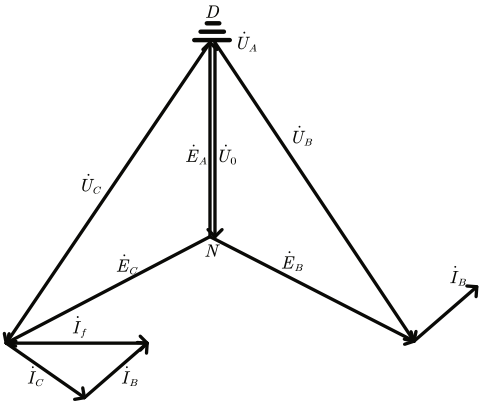


Fig. 2. The voltage and current vector diagram of an SLG fault on Line 2.

normal line 1, the phase-to-ground capacitive currents are given by: $\dot{I}_{A1} = 0, \dot{I}_{B1} = j\dot{U}_B\omega C_{D1}, \dot{I}_{C1} = j\dot{U}_C\omega C_{D1}$. In fault Line 1, the phase-to-ground capacitive currents are given by: $\dot{I}_{A2} = \dot{I}_f, \dot{I}_{B2} = j\dot{U}_B\omega C_{D2}, \dot{I}_{C2} = j\dot{U}_C\omega C_{D2}$. Due to the SLG fault occurring on phase A of Line 2, all the capacitive currents in the system flow through the grounding point towards phase A. Therefore, the fault current is given by: $\dot{I}_f = 3j\omega\dot{U}_0 C_\Sigma$. The zero-sequence current of the faulted Line 2 is given by Eq. (3). Based on Eqs. (2) and (3), the zero-sequence current of the normal line is the sum of the three-phase-to-ground capacitive currents flowing from the bus to the line. The zero-sequence current of the faulted line is the sum of the phase-to-ground capacitive currents of all the normal lines.

$$\dot{U}_0 = \frac{1}{3}(\dot{U}_A + \dot{U}_B + \dot{U}_C) = -\dot{E}_A \tag{1}$$

$$3\dot{I}_{01} = \dot{I}_{A1} + \dot{I}_{B1} + \dot{I}_{C1} = 3j\omega\dot{U}_0 C_{D1} \tag{2}$$

where, C_{D1} represents the phase-to-ground capacitance of Line 1.

$$3\dot{I}_{02} = \dot{I}_{A2} + \dot{I}_{B2} + \dot{I}_{C2} = 3j\omega\dot{U}_0\left(C_\Sigma - C_{D2}\right) \tag{3}$$

where, C_Σ represents the total system's phase-to-ground capacitance, and C_{D2} represents the phase-to-ground capacitance of Line 2.

When K is closed, it represents an arc suppression coil grounding system. During an SLG fault, the current at the inductance L is given by: $\dot{I}_L = \frac{\dot{U}_0}{j\omega L}$. The zero-sequence current of the normal Line 1 is the same as that of the neutral point ungrounded system, i.e., $3\dot{I}_L = j3\dot{U}_0\omega C_{D1}$. The fault current of the faulted Line 2 is as shown in Eq. (4). The arc suppression coil compensates for the phase-to-ground capacitive current, reducing the amplitude of the zero-sequence current in the faulted line, which lowers the accuracy of fault line selection.

$$\dot{I}_f = \dot{I}_L + \dot{I}_\Sigma = \frac{\dot{U}_0}{j\omega L} - 3j\omega C_\Sigma\dot{U}_0 \tag{4}$$

Based on the above analysis, faulty lines and normal lines show different characteristics during SLG fault. Therefore, identifying faulty lines can be regarded as a binary classification task (Gururajapathy et al., 2017).

2.2. Numerical spiking neural P system

To address the incompatibility between SNP systems and gradient descent-based training algorithms, the NSNP system was proposed (Wu et al., 2021). The NSNP system introduces variables and production functions, transforming spike transmission into numerical value transmission. An NSNP system is formally defined as a quadruple $\Pi = (N, syn, \sigma_{in}, \sigma_{out})$, where $N = \{\sigma_1, \sigma_2, \dots, \sigma_m\}$ denotes a finite set of neurons. Each neuron $\sigma_i = (V_i, P_i, V_i(0))$, for $1 \le i \le m$, comprises three components: a set of variables $V_i = \{x_{1,i}, x_{2,i}, \dots, x_{q,i}\}$, their corresponding initial values $V_i(0) = \{x_{1,i}(0), x_{2,i}(0), \dots, x_{q,i}(0)\}$, and a set of production functions P_i. The production functions in P_i are categorized into two types: non-threshold functions, denoted as $f_{n,i}(x_{1,i}, x_{2,i}, \dots, x_{q,i})$, and threshold functions, expressed as $f_{n,i}(x_{1,i}, x_{2,i}, \dots, x_{q,i})|_{T_n}$. The system also includes a set of synaptic connections $syn \subseteq \{(i,j) \mid 1 \le i, j \le m, i \ne j\}$, which establish links between neurons. Additionally, σ_{in} and σ_{out} designate the input and output neurons, respectively.

3. Memory dendritic neural (MDN) P system

Dendritic branches possess active information processing capabilities, enabling them not only to autonomously receive, integrate, and generate locally specific signals (such as "dendritic spikes") but also to perform differentiated computations based on input (Oh et al., 2025). The branching structure of dendrites forms a hierarchical information processing mechanism, in which distal dendrites serve as primary processing units that exhibit selective responses to specific input features, while proximal dendrites execute more complex signal integration functions (Xu et al., 2025). Drawing inspiration from these biological mechanisms, this section presents an MDNP system incorporating a dendritic hierarchical computational framework, elaborates its operational

mechanisms in detail, and illustrate the implementation of dendritic hierarchical computation through concrete examples.

3.1. Definition of the MDNP system

The MDNP system is represented by the following tuple:

$$\Pi = (N, \sigma_{in}, \sigma_{out}, syn)$$

(1) $N = \{\sigma_1, \ldots, \sigma_m\}$ is a set of m neurons, where neuron σ_j is in the form of $\sigma_j = (D_j, S_j)$, $1 \le j \le m$. Assume that the presynaptic neuron of σ_j is σ_i and the postsynaptic neuron is σ_h. The following is an introduction to the computational mechanism of σ_j:

1) $D_j = (d_{q,j}, dc)$ represents the dendrite of neuron σ_j, where $d_{q,j}$, $1 \le q \le n$ represents dendritic branches, and $dc \subseteq \left\{ (d_{q,j}, d_{k,j}) \middle| 1 \le q \ne k \le n \right\}$ represents dendritic connections. Each dendritic branch $d_{q,j}$ is defined as $d_{q,j} = (x_{q,j}, r_{q,j})$, where $x_{q,j}$ represents a numerical variable and $r_{q,j}$ is a rule that has the following two forms: a) Memory dendrite rules

$$f\left(mv_i(t-1), mv_{in}(t)\right) \xrightarrow[W_{q,j}, b_{q,j}]{} x_{k,j}(t)$$

Memory dendrite rules are used to transfer memory factors between connected dendritic branches. f is a production function (nonlinear function or polynomial function), $W_{q,j}$ is the dendritic weight, and $b_{q,j}$ is the dendritic bias value. $mv_i(t-1)$ and $mv_{in}(t)$ are the memory values transferred from the presynaptic neuron σ_i and the input neuron σ_{in} to the dendritic branch $d_{q,j}$ at time $t-1$ and time t, respectively. The computation result of $d_{q,j}$ is called the memory factor, which has different functions (such as forgetting, updating, etc.), expressed as $mf_{q,j}(t) = f\left(mv_i(t-1), mv_{in}(t-1)\right)$. Assume that at time t, there is a dendritic connection $(d_{q,j}, d_{k,j})$, then $d_{q,j}$ transmits the memory factor $mf_{q,j}(t)$ to the variable $x_{k,j}$ in the dendritic branch $d_{k,j}$.

b) Plasticity rules

$$mf_{q,j}(t) \xrightarrow{(d_{q,j}, d_{v,j})} x_{v,j}(t)$$

Plasticity rules regulate the connections of dendritic branches by disconnecting existing dendritic connections and creating new ones. $mf_{q,j}(t)$ is the memory factor of the dendritic branch $d_{q,j}$ at time t. Assume that at time t, $d_{q,j}$ executes the plasticty rule, then the existing dendritic connection $(d_{q,j}, d_{k,j})$ is disconnected and a new dendritic connection $(d_{q,j}, d_{v,j})$ is created, so $mf_{q,j}(t)$ is passed to the variable $x_{v,j}$ of the dendritic branch $d_{v,j}$ by the newly established dendritic connection $(d_{q,j}, d_{v,j})$.

2) $S_j = (s_j, r_j)$ represents the soma of neuron σ_j, where s_j is the numerical integration variable in the soma and r_j is the soma rule.

c) Soma rules

$$f\left(s_j(t)\right) \to \left[x_h(t), x_{out}(t)\right]$$

Soma rules are used to transfer the memory value of neuron σ_j to the postsynaptic neuron σ_h and the output neuron σ_{out}. Similarly, f is a production function (nonlinear function or polynomial function). Assuming that at time t, the soma S_j executes the soma rule, then the memory value $mv_j(t) = f(s_j(t))$ of neuron σ_j is transferred to the variables x_h and x_{out} in the dendrite branches of σ_h and σ_{out}, respectively.

(2) $syn = \left\{ (\sigma_i, \sigma_j) \middle| 1 \le i \ne j \le m \ OR \ i = in \ AND \ 1 \le j \le m \ OR \ 1 \le i \le m \ AND \ j = out \right\}$ represents the synaptic connection between neurons.

(3) σ_{in} and σ_{out} are the input and output neurons, respectively.

3.2. Operation mechanism of the MDNP system

The MDNP system consists of forward and reverse transmission neuron layers, where each neuron has dendritic branches as an independent computational unit. These branches perform computations through production functions (nonlinear or polynomial) under the regulation of memory dendritic rules or plasticity rules. Each branch can store multiple rules, but only one is activated in a single step, and the priority of memory dendritic rules is higher than that of plastic rules. The output of dendritic branches is memory factors, and the output of neurons is memory values.

The operation mechanism of the forward and reverse transmission of the MDNP system is similar. This section uses the forward transmission MDNP system as an example for detailed discussion, as shown in Fig. 3. Here, mv_{in}, mv_i, mv_j and mv_h are the memory values of neurons σ_{in}, σ_i, σ_j and σ_h at a certain moment, respectively. The dotted line indicates the dendritic connection to be created. The dendritic branch $d_{q,j}$ contains both the memory dendritic rule and the plasticity rule, while the branches $d_{k,j}$ and $d_{v,j}$ only contain the memory dendritic rule.

Assume that at time t, the dendrite D_j has a dendritic connection $(d_{q,j}, d_{k,j})$, and the dendritic branch $d_{q,j}$ preferentially executes the memory dendritic rule and transfers the memory factor $mf_{q,j}(t)$ to the dendritic branch $d_{k,j}$. At this time, the value of the variable $x_{k,j}$ is shown in Eq. (5). Then $d_{q,j}$ executes the plasticity rule to disconnect the dendritic connection $(d_{q,j}, d_{k,j})$ and create a new dendritic connection $(d_{q,j}, d_{v,j})$. The value of the variable $x_{v,j}$ is shown in Eq. (6).

$$x_{k,j}(t) = \begin{cases} mf_{q,j}(t) + x_{k,j}(t-1), & if \ (d_{q,j}, d_{k,j}) \in dc \\ x_{k,j}(t), & otherwise \end{cases} \tag{5}$$

$$x_{v,j}(t) = \begin{cases} mf_{q,j}(t) + x_{v,j}(t-1), & if \ (d_{q,j}, d_{v,j}) \in dc \\ x_{v,j}(t), & otherwise \end{cases} \tag{6}$$

where, $mf_{q,j}(t) = f\left(mv_i(t-1), mv_{in}(t-1)\right)$ is the memory factor of the dendritic branch $d_{q,j}$, $x_{k,j}(t-1)$ and $x_{v,j}(t-1)$ are the values of variables $x_{k,j}$ and $x_{v,j}$ at time $t-1$, respectively.

Assuming that at time t, the dendritic branches $d_{k,j}$ and $d_{v,j}$ are connected to the soma S_j, then the value of the variable s_j is shown in Eq. (7).

$$s_j(t) = \begin{cases} Concat\left[mf_{k,j}(t), mf_{v,j}(t)\right] + s_j(t-1), \\ if \ (d_{k,j}, S_j) \in dc \ and \ (d_{v,j}, S_j) \in dc \\ s_j(t-1), \qquad otherwise \end{cases} \tag{7}$$

where, $Concat\left[mf_{k,j}(t), mf_{v,j}(t)\right]$ is the concatenation of the memory factors produced by all dendritic branches connected to the soma S_j, and $s_j(t-1)$ is the value of the variable s_j at time $t-1$.

Neuron σ_j fires immediately after the value of variable s_j is updated. Specifically, in the MDNP system, each neuron not only connects with postsynaptic neurons but also establishes connections with the output neuron σ_{out}. Assuming that the postsynaptic neuron of σ_j is σ_h, then σ_j transmits the memory value $mv_j(t) = s_j(t)$ to neurons σ_h and σ_{out} through the synaptic channel. If the dendrite branch $d_{q,h}$ of σ_h receives the memory value $mv_j(t)$, then the value of the variable $x_{q,h}$ is shown in Eq. (8).

$$x_{q,h}(t) = \begin{cases} mv_j(t) + x_{q,h}(t-1), & if \ (\sigma_j, \sigma_h) \in syn \\ x_{q,h}(t-1), & otherwise \end{cases} \tag{8}$$

where $mv_j(t)$ is the memory value of neuron σ_j, $x_{q,h}(t-1)$ is the variable value of variable $x_{q,h}$ at time $t-1$. The situation of neuron σ_{out} is similar to that of σ_h, so it will not be repeated here.

The configuration of the MDNP system at time t is represented by the variable values of the dendrite branches of m neurons, i.e., $C_t = \left[x_{q,1}(t), x_{q,2}(t), \ldots, x_{q,m}(t)\right]$, $1 \le q \le n$, and the transition from one configuration to another is called a transformation. The computation of the system consists of a series of transformations, i.e., $C_0 \Rightarrow C_1 \Rightarrow C_2 \Rightarrow \cdots$.

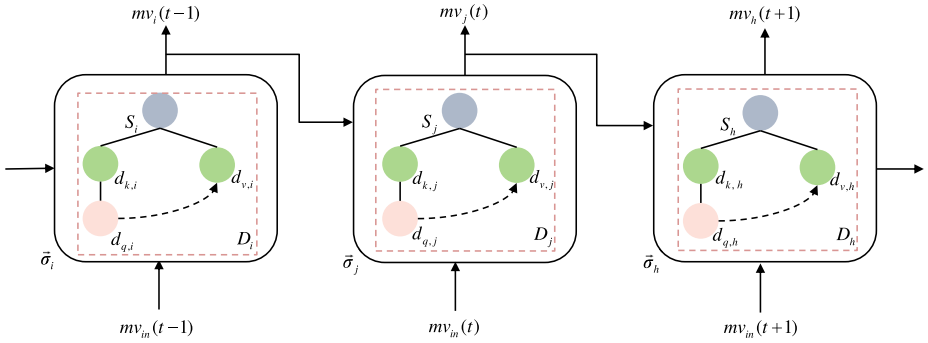


Fig. 3. An example of a forward-passing MDNP system Π_{ex}.

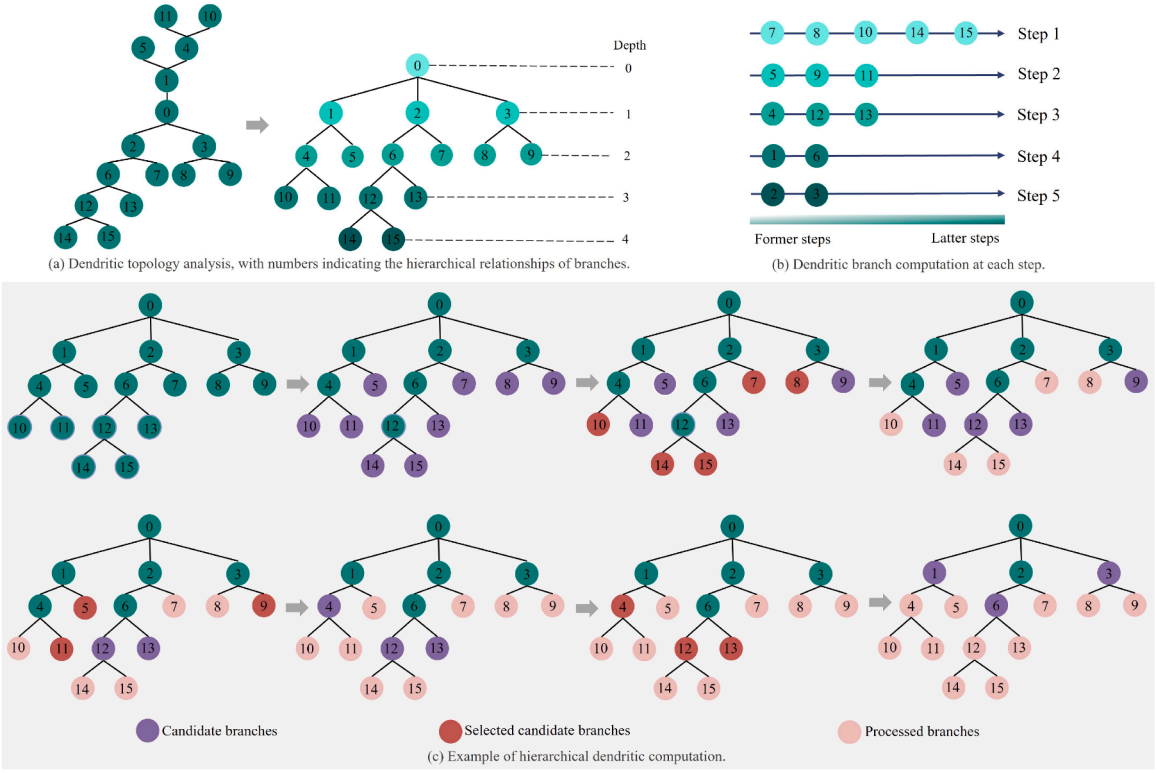


Fig. 4. Schematic illustration of hierarchical dendritic computation in the MDNP system. The dendritic branches are marked by numbered circular nodes. In the schematic diagram: (1) dark green nodes represent the initial nodes; (2) dark purple indicates candidate nodes that meet the computational criteria; (3) dark red nodes denote computational nodes randomly selected from the candidate pool; (4) pink nodes represent completed computational nodes.

3.3. Illustrative examples of dendritic hierarchical computation in MDNP systems

Fig. 4 illustrates the hierarchical computation process of dendritic branches in the MDNP system. The system represents each dendritic branch as a numbered circular node. Fig. 4(a) initially performs dendritic topology analysis to determine the connection relationships and depths of various branches. Fig. 4(b) distinguishes candidate branches (pending computation), selected branches (under computation), and processed branches (computation completed) using three distinct colors. In the first step, dendritic branches 5, 7, 9, 10, 11, 13, 14, and 15 await computation, yet only branches 7, 10, 14, and 15 are selected. The second step involves candidate branches 5, 9, 11, 12, and 13, with branches 5, 9, and 11 being chosen for computation. During the third step, candidate branches 4, 12, and 13 are selected for processing, where branch 13 applies plasticity rules to disconnect dendritic connection (13,6) while establishing connection (13,8). The fourth step identifies branches 1, 6, and 8 as candidates and selects them for computation. Subsequently, the fifth step designates branches 2 and 3 as candidate branches and executes computations. Finally, the computational results from branches 1, 2, and 3 are concatenated at the soma (represented by node 0), firing neuronal activation. Fig. 4(c) demonstrates the dendritic branches undergoing computation at each sequential step.

4. MDNP systems for multi-ground type fault line selection

This section proposes a fault line selection method based on the MDNP system for multiple grounding types. Its overall framework consists of three core modules: First, real-time operational data is obtained from the power scheduling system to construct the sample set, and data augmentation is performed using the CIM simulation model to address the issue of sample imbalance. Second, the neutral grounding type is determined based on CIM analysis, and the MDNP system is used to achieve high-accuracy fault line selection under different grounding types. Finally, an online application scheme is implemented to enable real-time training of the scheduling data, effectively overcoming the limitations of

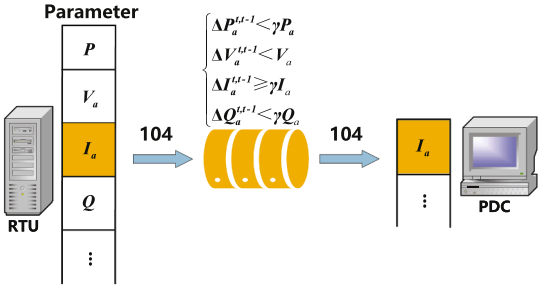


Fig. 5. Power scheduling data 104 protocol transmission rules.

Table 1
Real-time power scheduling data with a null value.

name	meritorious	reactivepower	I_a	I_b	I_c
10kVChing-Ma	0.42	−0.03	23.20	23.33	23.89
10kVChing-Ma	0.42	−0.03	Null	23.07	23.80
10kVChing-Ma	0.42	−0.03	23.37	23.07	23.80

traditional offline recording data in adapting to dynamic grid environments and enhancing the system's performance in practical operation.

4.1. Dataset construction

The raw data for the MDNP system primarily comes from the real-time scheduling data of the power operation control system (OCS), including active power (P), reactive power (Q), phase A current (I_A), phase B current (I_B), phase C current (I_C), and line voltage ($V_{a,b}, V_{b,c}, V_{c,a}$). For the SLG fault scenario, a fixed sliding time window is used to sample data for each line, covering the dynamic process before and after the fault occurs. Experimental data are collected over a 4-month period from real-time scheduling data, encompassing 29 neutral grounding systems with arc suppression coils and 94 neutral non-grounding systems.

In the data preprocessing stage, the issue of missing (null) real-time scheduling data values caused by the IEC60870-5-104 (referred to as 104) protocol (Egger et al., 2020), which affects the model's training performance, is addressed (Zhang et al., 2021). The transmission rules for power scheduling data are illustrated in Fig. 5. This paper uses linear interpolation to fill the missing values by calculating them based on the linear relationship between adjacent data points. The missing real-time scheduling data values are shown in Table 1. The formula for linear interpolation is given by Eq. (9). Finally, the neutral grounding through an arc suppression coil system provides 142 fault samples and 836 normal samples, while the neutral non-grounding system collects 278 fault samples and 1673 normal samples. To address the issue of sample imbalance, a simulation model based on CIM is established. Through simulation under different operating conditions, data augmentation is performed. The real-time data is then combined with the simulation data to construct a hybrid scheduling dataset, which serves as the input for the MDNP system.

$$M_v = M_{v0} + \frac{M_{v1} - M_{v0}}{x_1 - x_0}(x - x_0) \tag{9}$$

where M_v is the null value; M_{v1} and M_{v0} indicates the values of the data point to the left and right of the null value; x is the index of the null value; x_1 and x_0 denote left and right adjacent data point indexes.

4.2. MDNP systems for fault line selection

In this section, the neutral grounding type is determined based on CIM analysis, and the MDNP system is used for fault line selection under different grounding types.

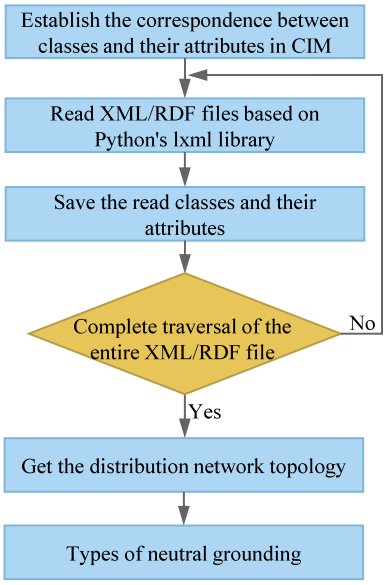


Fig. 6. CIM analysis for identifying the neutral grounding type.

Table 2
Relationship between classes and attributes in CIM.

Classes	Name	mRID	Terminals	ConnectivityNode
Substation	✓	✓	×	×
Power Transformer	✓	✓	✓	×
Transformer Winding	✓	✓	✓	×
Breaker	✓	✓	✓	×
Disconnector	✓	✓	✓	×
Ground Disconnector	✓	✓	✓	×
Busbar Section	✓	✓	✓	×
Energy Consumer	✓	✓	✓	×
AC Ligne Segment	✓	✓	✓	×
Compensator	✓	✓	✓	×
Terminal	✓	✓	×	✓
Connectivity Node	✓	✓	×	×

4.2.1. CIM-based analysis for neutral grounding type identification

In the CIM, XML/RDF files delineate the topological relationships among diverse power system components by defining class attributes and their inter-class associations. The main classes in CIM include Substation, Power Transformer, Transformer Winding, Disconnector, Breaker, Ground Disconnector, Busbar Section, Energy Consumer, AC Line Segment, Compensator, Connectivity Node, and Terminal. The attribute information mainly includes Name, mRID, Terminals, and ConnectivityNode. The relationship between classes and attributes in CIM is shown in Table 2. In Table 2, the first column specifies the equipment 'class,' followed by four columns representing distinct attributes. The symbol ✓ indicates that the current class possesses the attribute, while × indicates that the current class does not possess the attribute. Subsequently, the XML/RDF file is traversed to extract the information for each class. Based on these parsed results, the power system topology is constructed, through which the neutral grounding type is then identified. The detailed procedure for the CIM analysis is illustrated in Fig. 6.

4.2.2. MDNP systems for multi-grounding fault line selection

The MDNP system Π_{LS} addresses the low accuracy of fault line selection under different neutral grounding types by establishing neurons with dendritic hierarchical computation. As shown in Fig. 7, the system Π_{LS} employs a bidirectional information processing architecture consisting of two independent neuronal layers: forward and backward. The forward neuronal layer ($\vec{\sigma}_i$, $\vec{\sigma}_j$, $\vec{\sigma}_h$) processes data along the temporal direction to capture "past→future" dependencies, while the backward

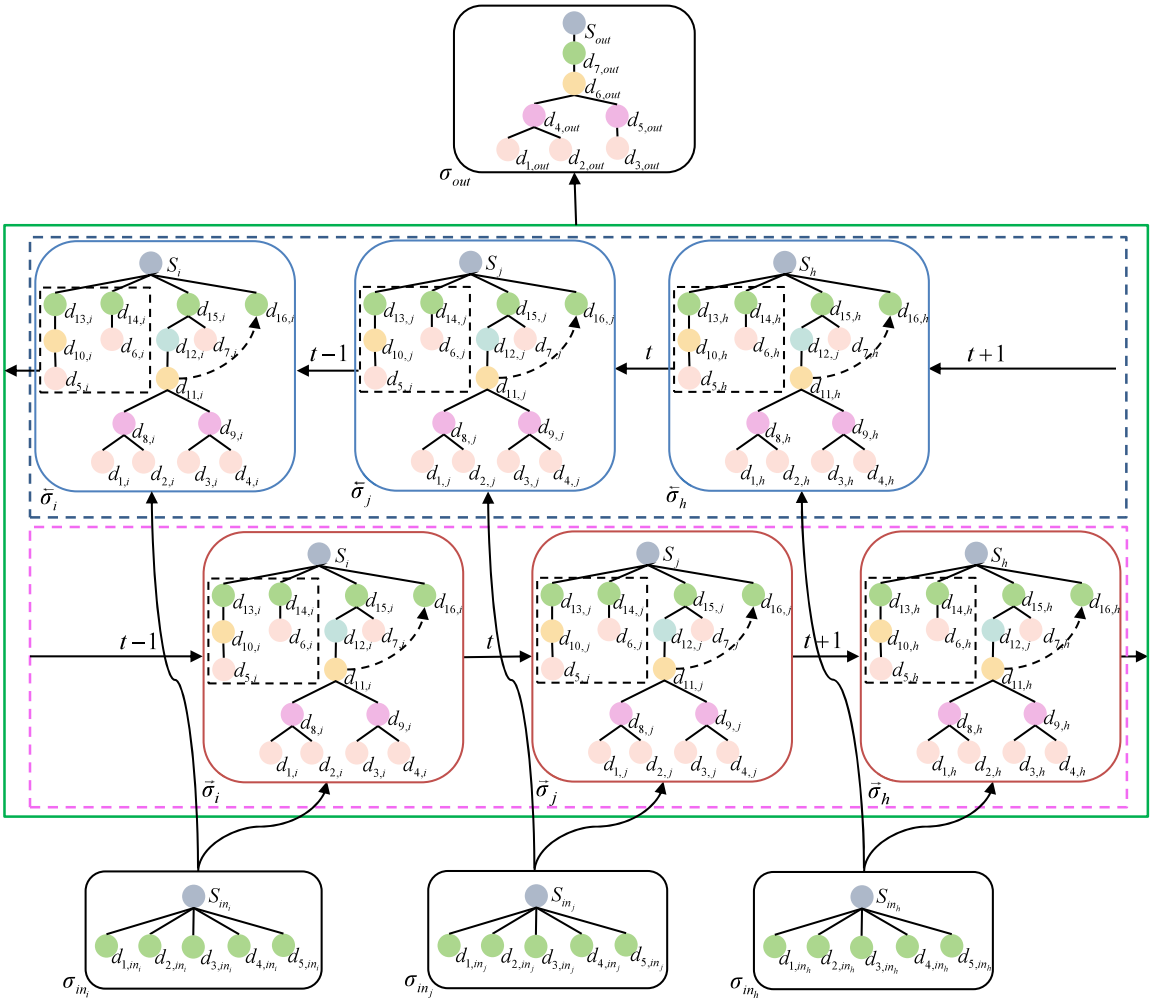


Fig. 7. The neuron structure of the MDNP system Π_{LS}.

neuronal layer ($\overleftarrow{\sigma}_i$, $\overleftarrow{\sigma}_j$, $\overleftarrow{\sigma}_h$) processes data in reverse temporal order to extract "future→past" dependencies. Through this bidirectional cooperative mechanism, the system can simultaneously model long-term dependencies in input sequence data, effectively integrating both historical and future fault characteristic information from power scheduling data along the temporal dimension, thereby significantly enhancing the representational capacity for sequential data.

The system Π_{LS} utilizes input neurons σ_{in} to concurrently feed power scheduling data into both forward and backward neuronal layers, ultimately concatenating the output feature vectors from both directions at the output neuron σ_{out}, thereby achieving bidirectional feature extraction and fusion. The system input comprises five electrical features: active power (P), reactive power (Q), and three-phase currents (I_a, I_b, I_c). This paper employs a sliding window approach for data preprocessing, represented as $X = (x_{1,\delta}, x_{2,\delta}, \ldots, x_{T,\delta})$, where each time step input $x_{\eta,\delta}$ contains the aforementioned five features and is delivered to the system through input neurons σ_{in}. The window size is configured as $T = 20$ with $1 \le \eta \le T$ and $1 \le \delta \le 5$. To eliminate dimensional discrepancies among features, the dendritic branches of σ_{in} normalize the input data at time t using Eq. (10), mapping all values to the interval [-1, 1].

$$\frac{2 * (x^l_{\eta,\delta}(t) - min(x^l_{\eta,\delta}(t))}{max(x^l_{\eta,\delta}(t)) - min(x^l_{\eta,\delta}(t))} - 1 \to s_{in} \tag{10}$$

where, $x^l_{\eta,\delta}(t)$ represents the δth feature value of the ηth sample in the original scheduling data at time t, where min and max denote the minimum and maximum values of $x^l_{\eta,\delta}(t)$, respectively. Consequently, the feature vector $\left[P^l_t, Q^l_t, I^l_{a,t}, I^l_{b,t}, I^l_{c,t}\right]$, composed of characteristic data from each line l, is transmitted to the variable s_{in} of the soma in σ_{in}.

In the system Π_{LS}, the dendritic branches of neurons dynamically adjust according to the neutral grounding configuration. For neutral non-grounding systems, the computational process primarily relies on dendritic branches $d_{1,j} - d_{4,j}$, $d_{7,j} - d_{9,j}$, $d_{11,j}$, $d_{12,j}$, $d_{15,j}$, and $d_{16,j}$. In contrast, for neutral grounding systems via arc suppression coils, where the inductive current generated by the coil partially compensates the capacitive current leading to weakened fault characteristics, additional dendritic branches $d_{5,j}$, $d_{6,j}$, $d_{10,j}$, $d_{13,j}$, and $d_{14,j}$ are incorporated to enhance feature extraction capability. At each time step t, the output of every neuron is transmitted not only to postsynaptic neurons but also to the output neuron.

A. Fault line selection of neutral non-grounding system

In neutral non-grounding systems, taking the forward-propagating neuron $\vec{\sigma}_j$ at time t as an example, This paper describe the computational process of system Π_{LS}. The input neuron σ_{in} transmits the feature vector $s_{in}(t) = \left[P^l(t), Q^l(t), I^l_a(t), I^l_b(t), I^l_c(t)\right]$ to dendritic branches $d_{1,j}$, $d_{2,j}$, $d_{3,j}$, $d_{4,j}$, and $d_{7,j}$ of $\vec{\sigma}_j$. At this stage, the input variables of all branches are initialized as $s_{in}(t)$, i.e., $x^l_{1,j}(t) = x^l_{2,j}(t) = x^l_{3,j}(t) = x^l_{4,j}(t) = x^l_{7,j}(t) = s_{in}(t)$. Simultaneously, the presynaptic neuron $\vec{\sigma}_i$ transmits the feature vector $s_i(t-1) = \left[o^l_i(t-1), c^l_i(t-1)\right]$ to the same dendritic branches $d_{1,j}$, $d_{2,j}$, $d_{3,j}$, $d_{4,j}$, and $d_{7,j}$, where $o^l_i(t-1)$ represents the output of neuron $\vec{\sigma}_i$ at time $t-1$, and $c^l_i(t-1)$ stores the state information.

The primary dendritic branches involved in the computation of neuron $\vec{\sigma}_j$ are $d_{1,j}$, $d_{2,j}$, $d_{3,j}$, $d_{4,j}$, and $d_{7,j}$. Among these, dendritic branch

$d_{1,j}$ transmits the state information of neuron $\vec{\sigma}_i$ at time $t-1$ to dendritic branch $d_{8,j}$ through Eq. (11).

$$c_i^l(t-1) \rightarrow x_{8,j}^l(t) \tag{11}$$

Dendritic branches $d_{2,j}$ and $d_{3,j}$ compute the forgetting memory factor $mf_{2,j}^l(t)$ and the updating memory factor $mf_{3,j}^l(t)$, respectively, through production function f_1, and transmit them to dendritic branches $d_{8,j}$ and $d_{9,j}$, as shown in Eqs. (12) and (13). Here, both $mf_{2,j}^l(t)$ and $mf_{3,j}^l(t)$ are constrained to the interval $[0,1]$.

$$mf_{2,j}^l = (f_1([o_i^l(t-1), x_{2,j}^l(t)]) \xrightarrow[W_{2,j}, b_{2,j}]{} x_{8,j}^l(t) \tag{12}$$

$$mf_{3,j}^l(t) = f_1([o_i^l(t-1), x_{3,j}^l(t)]) \xrightarrow[W_{3,j}, b_{3,j}]{} x_{9,j}^l(t) \tag{13}$$

where, $f_1(x) = \frac{1}{1+e^{-cx}}$. $W_{2,j}$, $W_{3,j}$, $b_{2,j}$ and $b_{3,j}$ represent learnable parameters.

The dendritic branch $d_{4,j}$ generates a candidate memory factor $mf_{4,j}^l(t)$ with gating functionality through the production function f_2 and transmits it to dendritic branch $d_{9,j}$, as specified in Eq. (14), where $mf_{4,j}^l(t)$ encapsulates the latent state information at time t.

$$mf_{4,j}^l(t) = f_2([o_i^l(t-1), x_{4,j}^l(t)]) \xrightarrow[W_{4,j}, b_{4,j}]{} x_{9,j}^l(t) \tag{14}$$

where, $f_2(x) = \tanh(x) = \frac{e^{2x}-1}{e^{2x}+1}$. $W_{4,j}$ and $b_{4,j}$ are learnable parameters.

The dendritic branch $d_{7,j}$ computes the memory factor $mf_{7,j}^l(t)$ via the production function f_1 and propagates it to dendritic branch $d_{15,j}$, as formalized in Eq. (15). This computation determines the output of neuron $\vec{\sigma}_j$ at time step t.

$$mf_{7,j}^l(t) = f_1([o_i^l(t-1), x_{7,j}^l(t)]) \xrightarrow[W_{7,j}, b_{7,j}]{} x_{15,j}^l(t) \tag{15}$$

where, $W_{7,j}$ and $b_{7,j}$ represent learnable parameters.

The second-level dendritic branches $d_{8,j}$ and $d_{9,j}$ perform the computation. Dendritic branch $d_{8,j}$ processes the state information $c_i^l(t-1)$ from neuron $\vec{\sigma}_i$ at time $t-1$ through the memory factor $mf_{2,j}^l(t)$ to filter historically irrelevant information, as formalized in Eq. (16).

$$c_i^l(t-1) * mf_{2,j}^l(t) \rightarrow x_{11,j}^l(t) \tag{16}$$

Dendritic branch $d_{9,j}$ regulates the retention of scheduling data features within the candidate memory factor $mf_{4,j}^l(t)$ through the updating-enabled memory factor $mf_{3,j}^l(t)$, with the specific computation detailed in Eq. (17).

$$mf_{3,j}^l(t) * mf_{4,j}^l(t) \rightarrow x_{11,j}^l(t) \tag{17}$$

The computational process of third-level dendritic branch $d_{11,j}$, as specified in Eq. (18), generates the state information $c_j^l(t)$ for neuron $\vec{\sigma}_j$ at time t, where $x_{8,j}^l(t)$ and $x_{9,j}^l(t)$ represent the feature matrices computed through Eqs. (12) and (13), respectively.

$$c_j^l(t) = x_{8,j}^l(t) + x_{9,j}^l(t) \rightarrow x_{12,j}^l(t) \tag{18}$$

The fourth-level dendritic branch $d_{12,j}$ applies the production function f_2 to smooth the state information $c_j^l(t)$, yielding the memory factor $mf_{12,j}^l(t)$ which is subsequently propagated to dendritic branch $d_{15,j}$, as mathematically formulated in Eq. (19).

$$mf_{12,j}^l(t) = f_2(c_j^l(t)) \rightarrow x_{15,j}^l(t) \tag{19}$$

where, $f_2(x) = \tanh(x) = \frac{e^{2x}-1}{e^{2x}+1}$.

The fifth-level dendritic branch $d_{11,j}$ undergoes plasticity rule, disconnecting from branch $d_{12,j}$ while forming a new connection with branch $d_{16,j}$, as described by Eq. (20).

$$c_j^l(t) \xrightarrow{(d_{11,j}, d_{16,j})} x_{16,j}^l(t) \tag{20}$$

The terminal-level dendritic branch $d_{15,j}$ computes the output $o_j^l(t)$ for neuron $\vec{\sigma}_j$ and transmits the result to the soma s_j, as formulated in Eq. (21).

$$o_j^l(t) = mf_{7,j}^l(t) \otimes mf_{12,j}^l(t) \rightarrow s_j(t) \tag{21}$$

where, $mf_{7,j}^l(t)$ and $mf_{12,j}^l(t)$ represent the memory factors of dendritic branches $d_{7,j}$ and $d_{12,j}$, respectively. The symbol $\otimes$ denotes matrix multiplication. The dendritic branch $d_{16,j}$ transmits the state information $c_j^l(t)$, computed by neuron $\vec{\sigma}_j$, to the soma S_j as specified in Eq. (22).

$$c_j^l(t) \rightarrow s_j(t) \tag{22}$$

The neuron $\vec{\sigma}_j$ fires upon updating its somatic state $s_j(t) = [o_j^l(t), c_j^l(t)]$ and simultaneously transmits the computed result $s_j(t)$ to both the output neuron σ_{out} and neuron $\vec{\sigma}_h$. Similarly, the backward-propagating neuron $\overleftarrow{\sigma}_j$ also transmits its computational result $\overleftarrow{s}_j(t) = [\overleftarrow{o}_j^l(t), \overleftarrow{c}j^l(t)]$ to both σ_{out} and $\overleftarrow{\sigma}_h$. Consequently, the feature matrices $\vec{s}_j(t)$ and $\overleftarrow{s}_j(t)$ from forward and backward computations are concatenated in the dendritic branch $d_{3,out}$ of σ_{out}, generating the memory factor $mf_{3,out}^l(t)$ which is then propagated to dendritic branch $d_{5,out}$, as formalized in Eq. (23).

$$mf_{3,out}^l(t) = f_3(\vec{s}_j(t), \overleftarrow{s}_j(t)) \rightarrow x_{5,out}^l(t) \tag{23}$$

where, f_3 denotes the $ConCat$ function.

The dendritic branch $d_{5,out}$ performs classification of line l according to Eq. (24) and propagates the classified memory factor $mf_{5,out}^l(t)$ to dendritic branch $d_{6,out}$, where Eq. (25) is executed to optimize $mf_{5,out}^l(t)$.

$$mf_{5,out}^l(t) = \frac{\exp(mf_{3,out}^l(t))}{\sum_{c=1}^{C} \exp(mf_{3,out}^l(t))} \rightarrow x_{6,out}^l(t) \tag{24}$$

$$mf_{6,out}^l(t) = -\frac{1}{N} \sum_{n=1}^{N} \sum_{c=1}^{C} mf_{5,out}^{l,n}(t) \log mf_{5,out}^{l,n}(t) \rightarrow x_{5,out}^l(t) \tag{25}$$

where, C represents the number of classification categories and N denotes the total number of training samples.

During the training process, if the loss memory factor $mf_{6,out}^l(t)$ remains unchanged over consecutive iterations, an adaptive learning mechanism is employed to accelerate the convergence of the MDNP system, as specified in Eq. (26).

$$\theta_t - lr_{init} \cdot factor \cdot \nabla mf_{6,out}^l(\theta_t) \rightarrow s_{out}(t) \tag{26}$$

where lr_{init} is the initial learning parameter setting value, $factor$ is the learning parameter adjustment factor, and θ represents the learning weight parameter of the MDNP system. The MDNP system terminates optimization upon reaching the predetermined iteration count. The soma S_{out} executes $s_{out}(t) \rightarrow evn$ to transmit the classification results to the external environment, where $s_{out}(t) = mf_{5,out}^l(t)$ and evn denotes the external environment.

B. Fault line selection of neutral arc suppression coil system

For the neutral by arc suppression coil grounding system, this paper incorporates additional dendritic branches $d_{5,j}$, $d_{6,j}$, $d_{10,j}$, $d_{13,j}$, and $d_{14,j}$ for each neuron in system Π_{LS}. The first-level dendritic branches $d_{5,j}$ and $d_{6,j}$ initiate the computation process. Specifically, $d_{5,j}$ calculates the maximum three-phase current variation, generates the memory factor $mf_{5,j}^l(t)$, and transmits it to dendritic branch $d_{10,j}$, as formulated in Eq. (27).

$$mf_{5,j}^l(t) = \max\{I_a^l(t) - I_a^l(t-1), I_b^l(t) - I_b^l(t-1), I_c^l(t) - I_c^l(t-1)\} \rightarrow x_{10,j}^l(t) \tag{27}$$

The dendritic branch $d_{6,j}$ computes the reactive power variation, generates the memory factor $mf_{6,j}^l(t)$, and transimits it to dendritic branch $d_{14,j}$, as specified in Eq. (28).

$$mf_{6,j}^l(t) = Q^l(t) - Q^l(t-1) \rightarrow x_{14,j}^l(t) \tag{28}$$

The third-level dendritic branch $d_{10,j}$ performs Z-score normalization on the memory factor $mf_{5,j}^l(t)$ and transmits the result $mf_{10,j}^l(t)$ to dendritic branch $d_{1,j}$, as formulated in Eq. (29).

$$mf_{10,j}^l(t) = \frac{mf_{5,j}^l(t) - \mu}{\sigma} \rightarrow x_{1,j}^l(t) \tag{29}$$

where, μ and σ denote the mean and standard deviation of $mf^{l}_{5,j}(t)$ across all lines, respectively.

In the final layer, dendritic branches $d_{13,j}$ and $d_{14,j}$ compute the normalized values $\alpha^{l}_{j}(t)$ of the memory factor $mf^{l}_{10,j}(t)$ and $\beta^{l}_{j}(t)$ of memory factor $mf^{l}_{6,j}(t)$ for the lth line, respectively, and transmit these computational results to the soma S_j, as described in Eqs. (30) and (31).

$$\alpha^{l}_{j}(t) = \frac{mf^{l}_{10,j}(t) - \min mf^{l}_{10,j}(t)}{\min mf^{l}_{10,j}(t) - \min mf^{l}_{10,j}(t)} \rightarrow s_j(t) \tag{30}$$

$$\beta^{l}_{j}(t) = \frac{mf^{l}_{6,j}(t) - \min mf^{l}_{6,j}(t)}{\min mf^{l}_{6,j}(t) - \min mf^{l}_{6,j}(t)} \rightarrow s_j(t) \tag{31}$$

At this stage, the somatic state S_j updates to $s_j(t) = \left[\vec{\alpha}^{l}_{j}(t), \vec{\beta}^{l}_{j}(t), \vec{o}^{l}_{j}(t), \vec{c}^{l}_{j}(t)\right]$, triggering the activation of neuron $\vec{\sigma}_j$.

Similarly, at time t, the forward-propagating neuron $\vec{\sigma}_j$ and the backward-propagating neuron $\overleftarrow{\sigma}_j$ simultaneously transmit the calculation results to the dendritic branches $d_{1,out}, d_{2,out}, d_{3,out}$ of σ_{out}. Each dendritic branch concatenates the feature vectors obtained from bidirectional computations and subsequently relays the memory factors $mf^{l}_{1,out}(t)$, $mf^{l}_{2,out}(t)$, and $mf^{l}_{3,out}(t)$ to downstream branches $d_{4,out}$ and $d_{5,out}$, as mathematically formulated in Eqs. (32)–(34).

$$mf^{l}_{1,out}(t) = f_3(\vec{\alpha}^{l}_{j}(t), \overleftarrow{\alpha}^{l}_{j}(t)) \rightarrow x^{l}_{4,out}(t) \tag{32}$$

$$mf^{l}_{2,out}(t) = f_3(\vec{\beta}^{l}_{j}(t), \overleftarrow{\beta}^{l}_{j}(t)) \rightarrow x^{l}_{4,out}(t) \tag{33}$$

$$mf^{l}_{3,out}(t) = f_3(\vec{o}^{l}_{j}(t), \overleftarrow{o}^{l}_{j}(t)) \rightarrow x^{l}_{5,out}(t) \tag{34}$$

where, f_3 represents the $ConCat$ function.

The dendritic branches $d_{4,out}$ and $d_{5,out}$ execute Eqs. (35) and (36) respectively, transmitting the fault probabilities $mf^{l}_{4,out}(t)$ and $mf^{l}_{5,out}(t)$ to the dendritic branch $d_{6,out}$. Subsequently, the dendritic branch $d_{6,out}$ executes Eq. (25) to optimize the memory factor $mf^{l}_{5,out}(t)$, while $d_{7,out}$ implements Eq. (26) to perfom an adaptive learning mechanism that accelerates convergence of the MDNP system. The MDNP system terminates the optimization process upon reaching the predetermined iteration threshold.

$$mf^{l}_{4,out}(t) = \lambda mf^{l}_{1,out}(t) + (1-\lambda) mf^{l}_{2,out}(t) \rightarrow x^{l}_{6,out}(t) \tag{35}$$

$$mf^{l}_{5,out}(t) = \frac{\exp(mf^{l}_{3,out}(t))}{\sum_{c=1}^{C} \exp(mf^{l}_{3,out}(t))} \rightarrow x^{l}_{6,out}(t) \tag{36}$$

where, λ denotes the fault weighting coefficient. C is the number of classification categories.

Within the soma S_{out}, the output of neuron σ_{out} undergoes a fault detection strategy: if $\min(mf^{l}_{4,out}(t), mf^{l}_{5,out}(t)) > \zeta_1$ is satisfied, the line is classified as faulty. Otherwise, the system evaluates $mf^{l}_{4,out}(t) > \zeta_2$, if true, the line is determined to be faulty. If neither condition holds, the system checks $mf^{l}_{5,out}(t) > \zeta_3$, classifying the line as faulty when met, otherwise designating it as normal, as mathematically defined in Eq. (37).

$$f_4[mf^{l}_{4,out}(t), mf^{l}_{5,out}(t)] \rightarrow env \tag{37}$$

where

$$f_4 = \begin{cases} 1 & \text{if } \min(mf^{l}_{4,out}(t), mf^{l}_{5,out}(t)) > \zeta_1 \text{ or } mf^{l}_{4,out}(t) > \zeta_2 \\ & \text{or } mf^{l}_{5,out}(t) > \zeta_3 \\ 0 & \text{otherwise} \end{cases}$$

, and the decision coefficients ξ_1, ξ_2, and ξ_3 are typically set as constants within the range $[0, 1]$.

4.2.3. Online application

In this section, the offline trained MDNP system is deployed to the actual distribution network to realize real-time fault line selection. The system is trained online to optimize the parameters by collecting the operating data of the faulty line before and after the fault. In parallel, real-time data are used to continuously test and validate the system's performance. The block diagram of the online application of the MDNP system is shown in Fig. 8. The following provides a detailed description of the online application process.

Step 1: Analyze CIM to generate the power topology of the distribution network.

In the power scheduling system center, the CIM model file is used to store the power topology of the distribution network. By analyzing the CIM model, the detailed power topology of the distribution network is derived. This paper analyzes the relationships and information between classes by parsing the XML/RDF format CIM model, thereby extracting the topology relationship diagram of the distribution network.

Step 2: Real-time scheduling data collection and processing.

The power topology of the distribution network obtained in Step 1 is used to read real-time scheduling data. Data preprocessing is performed according to Eq. (10). The real-time scheduling data primarily includes: $P, Q, I_a, I_b, I_c, V_{ab}, V_{bc}$ and V_{ca}.

Step 3: Fault line selection start-up criteria.

Monitor the line voltage signal at the bus node (V_{ab}, V_{bc}, V_{ca}) to determine whether the fluctuation range exceeds the set threshold. During an SLG fault, the line voltage at the bus undergoes significant changes. Based on practical engineering experience, this paper selects a 10% change in line voltage as the start-up criterion for fault line selection method.

Step 4: Output of results.

Real-time scheduling data serves as the input to the MDNP system. Through dendritic hierarchical computation within the MDNP system, the final output is the result of fault line selection. For the lines in the distribution network that need to be predicted, multiple prediction results can be obtained, represented as $Y = (Y_1, Y_2, \ldots, Y_l)$. The line with the highest probability of failure is comprehensively judged to be the faulty line.

Step 5: Saving results and online training.

After each fault line selection, the fault log in the scheduling and line selection system will automatically record the faulty line and dynamically expand the real-time fault data to the training data set of the MDNP system. As the SLG fault line selection system continues to operate, the dataset gradually expands, and the MDNP system parameters are autonomously updated and optimized through online training.

5. Experimental results

5.1. Description of the experimental dataset

The hybrid scheduling dataset constructed in this paper served as the input data source for the MDNP system. Its simulation data was generated through a CIM-based simulation model. The relevant experimental details were described in this section. The 10 kV/35 kV distribution network simulation model established based on CIM is shown in Fig. 9. The simulation topology dynamically adjusts the topology based on circuit breakers, enabling the acquisition of simulation data for different topologies. The configuration of the simulation model adopts a three-phase infinite power supply system with a rated capacity of 300 MV·A, and the transformer T has a capacity of 31.5 MV·A. T_z has been a grounded transformer of type Z, constructed with a neutral point, where 'J' indicates an overhead feeder and 'D' indicates a cable feeder. The arc suppression coil employs an overcompensated configuration to render the grounding current inductive, preventing resonant overvoltage. The compensation level is set to the standard 10% compensation value commonly adopted in most practical applications (Wang & Cui, 2022). The resistance of the arc suppression coil has been taken as 3%

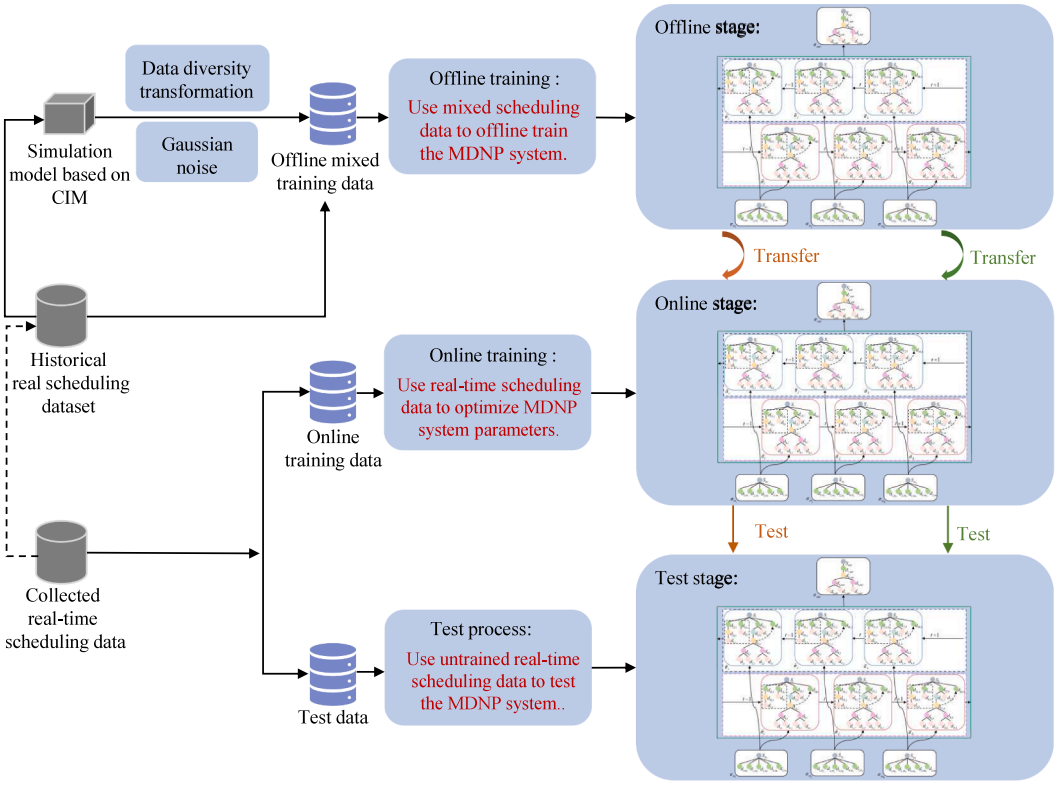


Fig. 8. Block diagram of the MDNP system for online application.

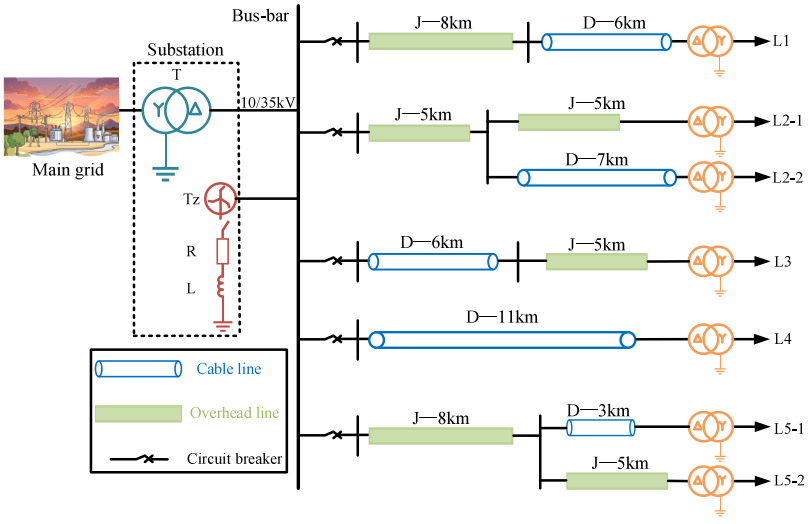


Fig. 9. Simulation model of 10/35-kV distribution network.

of the inductive resistance, with the measured value of $R = 5.26\,\Omega$. The parameter configurations of the remaining feeders are summarized in Table 3. The inductance calculation of the arc suppression coil is shown in Eq. (38).

$$L = \frac{1}{3\omega^2 C_{\Sigma}(1+10\%)} \tag{38}$$

where C_{Σ} denotes the aggregate ground capacitance across all feeders, with an inductance value of $L = 0.9235$ H determined through calculation.

To increase the diversity of the data samples, this paper simulated SLG faults in the distribution network under different operating conditions, considering the impact of key parameters such as line type (LT), fault resistance (FR), fault distance (FD), and fault type (FT). In addition, to simulate the impact of environmental noise on fault charac-

Table 3
Feeder parameters.

Feeder types	Phase-sequence	R (Ω/km)	L (mH/km)	C (μF/km)
Overhead feeder	Zero-sequence	0.2300	5.4780	0.0067
	Positive-sequence	0.1700	1.2100	0.0127
Cable feeder	Zero-sequence	0.1965	3.1050	0.1400
	Positive-sequence	0.0242	1.6440	0.5500

Table 4
Parameters of the simulation data.

Parameter	Contents
Line types (LT)	Cable Line, Overhead Line, Hybrid Line
Fault resistance (FR)	10Ω, 50Ω, 100Ω, 200Ω, 300Ω, 400Ω, 500Ω, 600Ω, 700Ω, 800Ω, 900Ω, 1000Ω, 1200Ω, 1400Ω, 1500Ω
Fault distance (FD)	1km, 2km, 3km, 5km, 6km, 8km, 10km, 12km, 14km
Faulty types (FT)	Typical grounding fault, Arc grounding fault
Noise ratio (SNR)	1dB, 10dB, 15dB, 20dB

Table 5
Statistics of the hybrid scheduling dataset.

Different grounding types	Hybrid scheduling dataset	
	Fault	Noraml
Arc suppression coil	900	2661
Non-grounding	1130	3099
Total	2030	5760

Table 6
Parameter settings of the MDNP system.

Parameter name	Symbol	Value
Initial learning rate	lr_{init}	0.001
Factor	f	0.35
Batch size	b	64
Max epoch	e	250
Dropout rate	dr	0.3
Optimiser	adam	/
Loss function	cross_entropy	/
Fault weight coefficient	λ	0.52
Decision coefficients	$\zeta_1, \zeta_2, \zeta_3$	0.5,0.8,0.9

teristics, gaussian white noise of 1 dB, 10 dB, 15 dB, and 20 dB was injected into the three-phase current (I_a, I_b, I_c) and reactive power (Q) characteristic data, respectively. The specific simulation parameter configuration is detailed in Table 4.

After the experiments, the simulation data samples include 758 fault samples and 1825 normal samples in the neutral with arc suppression coil, and 852 fault samples and 1426 normal samples in the neutral non-grounding system. This paper constructed a hybrid scheduling dataset by combining the simulation data with real-time scheduling data in a 1:3 ratio, resulting in 7790 samples, including 2030 fault samples and 5760 normal samples. The sample statistics of the hybrid scheduling dataset are shown in Table 5.

5.2. Implementation details

The MDNP system was experimented on the Pytorch framework and Matlab software, and all the experiments involved were implemented on GPU 3090, 64GB RAM. The parameter settings of the MDNP system are shown in Table 6.

The hybrid scheduling dataset obtained in this paper contains a total of 7680 data samples, which are used for offline training and validation under different neutral grounding methods in a 9:1 ratio. Additionally, the MDNP system is applied to online training and validation in an actual distribution network. Accuracy (ACC), recall (REC), and precision (PRE) are selected as evaluation metrics for the MDNP system. The MDNP system consists of one input neuron, one output neuron, and 256 neurons in the forward and backward layers.

5.3. Verification of consistency between real-time scheduling data and simulation data

This section aims to verify the consistency between the simulation data and real-time scheduling data in the hybrid scheduling dataset. To achieve this, 500, 1000, 1500, 2000, and 2500 random data samples were selected from the simulation data, and consistency analysis was performed using statistical data analysis methods. The mean and variance statistics of each dataset are shown in Table 7. As can be seen from the comparison, the mean and variance of each feature parameter in the samples were relatively small, indicating that the simulation data generated based on CIM was highly consistent with the real-time scheduling data.

Table 7
Consistency analysis of data samples.

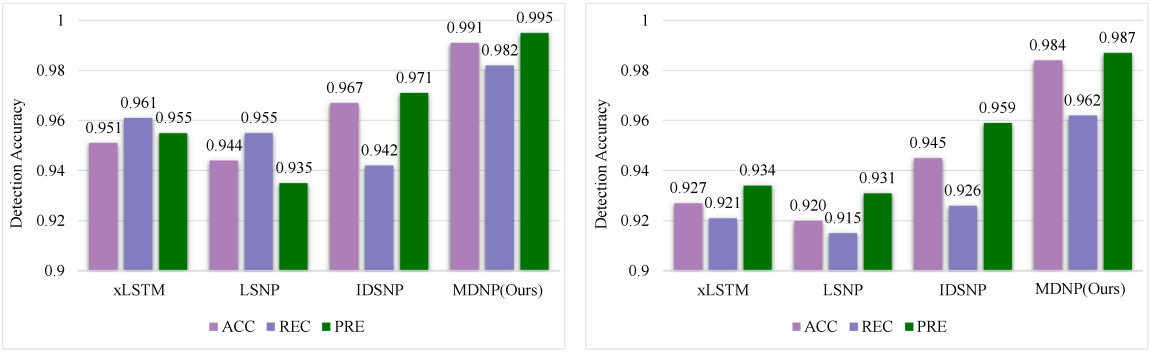
Training dataset sizes	Mean/Variance (%)				
	P	Q	I_a	I_b	I_c
500	0.32/1.13	2.15/1.36	1.21/4.32	1.69/0.87	1.42/2.53
1000	0.64/3.42	1.23/2.29	1.84/4.29	1.07/0.56	0.55/1.94
1500	0.29/3.15	0.56/1.11	0.44/1.58	0.91/2.38	1.54/1.13
2000	0.25/2.71	0.95/1.57	0.31/1.20	0.55/1.33	0.67/1.89
2500	0.20/1.56	0.46/1.10	0.22/0.96	0.49/0.42	0.58/1.17

5.4. Performance comparison with other methods on the hybrid scheduling dataset

This paper verifies the fault line selection performance of the MDNP system under different grounding types by selecting eight baseline models for comparative analysis. Additionally, model training and testing evaluations were conducted using training sample sets of varying sizes. The experimental results showed that, under both the neutral non-grounding and arc suppression coil grounding types, the MDNP system demonstrated significant performance advantages. As shown in Table 8, with the increase in the training sample size, the ACC of the MDNP system showed a continuous improvement trend. When the training sample size reached 7000, the ACC of the MDNP system under the neutral point ungrounded and arc suppression coil grounded types achieved 99.1% and 98.4%, respectively. As shown in Fig. 10, compared to the xLSTM, LSNP and IDSNP systems, the MDNP system demonstrated significant advantages in the three metrics of ACC, REC, and PRE. Specifically, the ACC increased by as much as 6.4%, REC improved by 4.7%, and PRE rose by 6%, fully validating the superiority of the proposed method.

In particular, compared to the IDSNP system proposed in previous work, the MDNP system designed in this paper significantly improved the fault line selection accuracy under different grounding types by introducing a triggering rule with memory dendritic factors and a dynamically adjustable dendritic branching structure. To address the weak SLG fault features with neutral grounding via an arc suppression coil, the MDNP system optimized dendritic branch configuration, enhancing its feature extraction capability. Additionally, based on the biological neuronal dendritic computation mechanism, the MDNP system innovatively adopted a hierarchical computing architecture, reducing neuronal resource consumption. Specifically, the IDSNP system contains 590 neurons (including 16 gated neurons, 48 rule neurons, and 526 weighted neurons), while the MDNP system achieves superior fault line selection performance with 256 neurons (forward and backward layers neurons), resulting in a 56.6% reduction in computational resource consumption. This structural optimization and computational approach enabled the MDNP system to demonstrate greater engineering application value in terms of both accuracy and efficiency.

The comparison of training and testing time across varying dataset scales is summarized in Table 9. The experimental results indicate that while traditional machine learning methods (e.g., AdaBoost, SVM) incur lower computational overhead, their accuracy falls below 80%, rendering them inadequate for fault line selection requirements. In contrast, the MDNP system demonstrates a significant computational advantage while maintaining high precision. At a sample size of 7000, its training time (203.0s) was approximately 24.5% and 26% shorter than that of xLSTM (269.1s) and Swin-Transformer (S-T) (274.5s), respectively, with a stable testing time of 0.15s. Although the training duration of MDNP is



(a) Comparison of ACC, REC, PRE values in the neutral non-grounding type.

(b) Comparison of ACC, REC, PRE values in the neutral arc suppression coil type.

Fig. 10. Comparison of ACC, REC, and PRE between the MDNP system and other SNP systems under different grounding types, as shown in (a) and (b).

Table 8
Fault line Selection Accuracy of different methods on training datasets of different scales under different grounding types (Arc suppression coil/Non-grounding).

Methods	Detection accuracy (Arc suppression coil/Non-grounding)			
	1000	3000	5000	7000
Adaboost (Su et al., 2022)	0.564/0.594	0.628/0.683	0.705/0.723	0.776/0.792
SVM (Xiong et al., 2024)	0.534/0.582	0.598/0.615	0.685/0.705	0.753/0.775
LSTM (Yuan & Jiao, 2023)	0.638/0.647	0.742/0.784	0.821/0.845	0.841/0.852
Bi-LSTM (Su et al., 2021)	0.623/0.695	0.756/0.816	0.859/0.874	0.872/0.895
xLSTM (Dong et al., 2025)	0.642/0.713	0.771/0.827	0.903/0.922	0.927/0.951
S-T (Guo et al., 2025b)	0.635/0.704	0.790/0.831	0.877/0.910	0.914/0.929
LSNP (Hu et al., 2024)	0.605/0.645	0.762/0.820	0.884/0.915	0.920/0.944
IDSNP (Ma et al., 2024)	0.627/0.710	0.781/0.850	0.910/0.946	0.945/0.967
MDNP(Ours)	**0.705/0.791**	**0.839/0.892**	**0.965/0.965**	**0.984/0.991**

Table 9
Comparison of training and test times for different methods on training datasets of differing scales.

Methods	Training time(s)/Test time(s)			
	1000	3000	5000	7000
Adaboost (Su et al., 2022)	0.21/0.02	0.39/0.03	0.67/0.02	0.89/0.03
SVM (Xiong et al., 2024)	0.09/0.02	0.23/0.02	0.46/0.03	0.65/0.04
LSTM (Yuan & Jiao, 2023)	74.2/0.12	91.1/0.11	124.7/0.11	151.9/0.12
Bi-LSTM (Su et al., 2021)	123.8/0.14	167.5/0.13	194.5/0.15	238.7/0.15
xLSTM (Dong et al., 2025)	155.4/0.15	180.3/0.14	233.2/0.15	269.1/0.14
S-T (Guo et al., 2025b)	143.7/0.14	184.1/0.16	251.3/0.17	274.5/0.16
LSNP (Hu et al., 2024)	95.2/0.13	122.4/0.11	141.7/0.12	192.3/0.12
IDSNP (Ma et al., 2024)	127.9/0.14	149.0/0.15	173.1/0.17	216.5/0.16
MDNP(Ours)	**117.5/0.12**	**160.2/0.13**	**185.8/0.15**	**203.0/0.15**

slightly higher than that of the LSNP system (192.3s), it offers superior selection accuracy. In summary, the MDNP system effectively balances computational efficiency with high accuracy performance, exhibiting robust convergence capabilities and inference efficiency in large-scale data processing.

5.5. Evaluation of fault line selection accuracy under high-impedance ground fault data and noise data

This section utilizes a hybrid dataset consisting of real-time scheduling data and simulated data (7790 samples) for model training. The high-impedance ground fault data were constructed using the Emanuel model, and the Gaussian white noise in the noise data was configured to be 1~20 dB. The following is the model performance test under high-impedance grounding fault and noise influence.

(1) Performance testing under the influence of high-impedance grounding fault

Table 10
Distribution of sample data for high-impedance grounding fault.

Neutral grounding types	Arc model	Fault line	Fault phase	Fault initial phase angle/°	Fault distance/km	data sample
Arc suppression coil Non-grounding	Emanual	L1~L5	A~C	0° ~ 180°	2km~14km	940 618

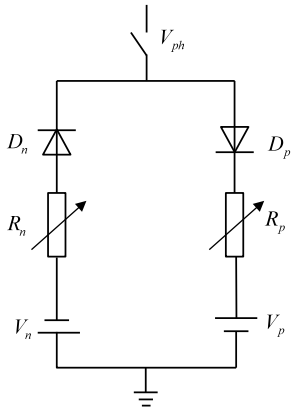


Fig. 11. Emanuel model.

Based on the Emanuel model, this paper establishes an arc simulation model comprising two sets of reverse-parallel DC voltage sources, diodes, and resistive (Bai et al., 2022). This structure is simple and effectively characterizes the nonlinear characteristics of arc resistance, as shown in Fig. 11. The two DC voltage sources are $V_p = 3500$V and $V_n = 3000$V, with variable resistors R_p and R_n was set ranging from 500Ω to 3000Ω. V_p represents the phase voltage. The construction of a high-impedance grounding fault data sample set was achieved by employing the arc simulation model to replicate operational scenarios of high-impedance grounding faults occurring under different conditions within actual distribution networks. The fault parameter settings are detailed in Table 10. There are a total of 2337 high-impedance ground fault data samples under two different grounding configurations, accounting for 30% of the total hybrid scheduling dataset. Specifically, the high-impedance ground fault data samples under neutral grounding via arc suppression coil and non-grounding types were 1402 and 935, respectively. To validate the fault line selection performance of the proposed MDNP system, test samples comprising 180 and 150 instances were constructed, respectively. The test results in Fig. 12 demonstrate that the MDNP system exhibits good high-impedance ground fault identification capability under two grounding configurations, achieving accuracy of 0.956 and 0.967, respectively.

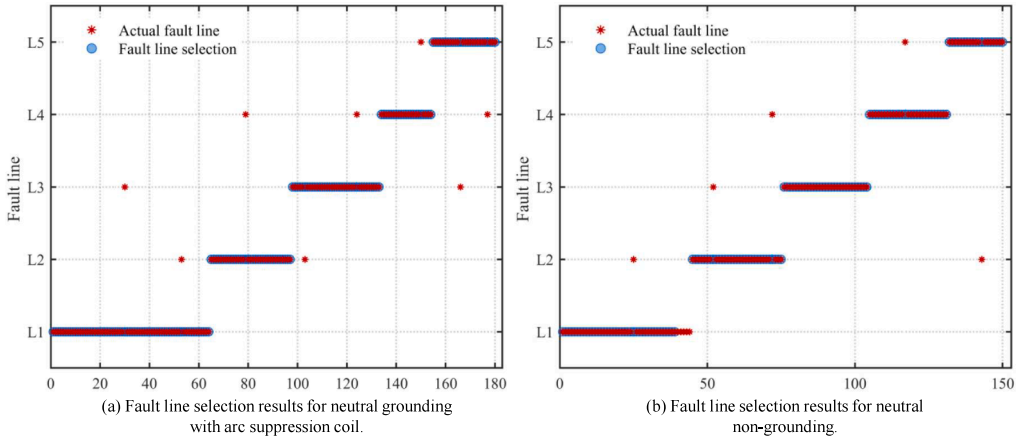


Fig. 12. Result of high-impedance grounding fault testing for the MDNP system under different grounding types.

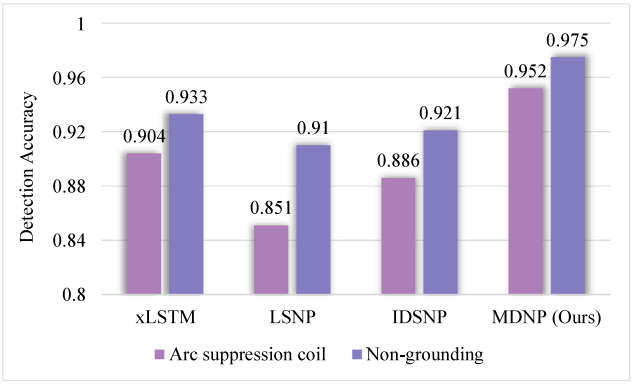


Fig. 13. Comparison of fault line selection accuracy under noise influence for different models.

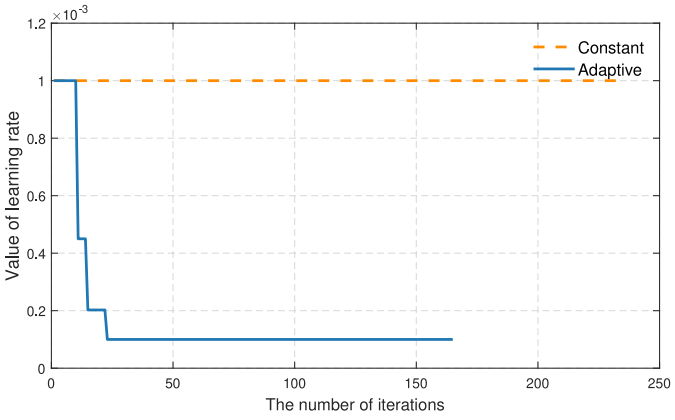


Fig. 14. Variation of iterative loss in the MDNP system with constant and adaptive learning rate.

(2) Performance testing under noise influence

To validate the model's fault line selection accuracy under noise interference, this paper constructed a noise dataset by introducing 1–20 dB Gaussian white noise. Under the neutral grounding with arc suppression coil, a total of 1050 samples were generated (comprising 360 fault samples and 690 normal samples); under the neutral non-grounding configuration, 850 samples were generated (comprising 280 fault samples and 570 normal samples). The comparison of fault line selection accuracy under noise influence for different models is shown in Fig. 13. The experimental results demonstrate that under noisy conditions, the proposed MDNP model exhibits significant advantages, achieving detection accuracies of 0.952 and 0.975 for grounding with arc suppression coils and non-grounding configurations, respectively, both markedly higher than the xLSTM, LSNP, and IDSNP models. MDNP not only ensures high fault line selection accuracy under different grounding configurations but also effectively mitigates noise interference in feature extraction, fully validating the model's outstanding noise resistance in distribution network fault line selection tasks.

5.6. Ablation study

To evaluate the impact of the adaptive learning mechanism on the fault line selection performance of the MDNP system, this paper compared and analyzed the experimental results of the constant learning rate and the adaptive learning rate. As shown in Figs. 14 and 15, with a training sample size of 7000, the fault and normal samples under the configuration of the neutral grounding via arc suppression coil were 808 and 2392, respectively, while those under the non-grounding configuration were 1015 and 2785. The MDNP system with an adaptive learning rate achieved an increase in fault line selection accuracy of 1.40% and 1.20% for the arc suppression coil grounding and the non-grounding types, respectively. In addition, the convergence curves in Fig. 16 show that after introducing the adaptive learning mechanism, the number of iterations required for convergence decreased from 233 to 165, significantly accelerating the model's convergence speed and further validating the effectiveness of this optimization strategy.

5.7. Evaluation of online application

To verify the online fault line selection performance of the MDNP system in actual distribution networks, real-time tests were conducted under two types of neutral grounding types (arc suppression coil grounding and non-grounding) in a distribution network in Guangxi. The distribution quantities for the two grounding types are 29 and 94, respectively, as shown in Table 11. The fault logs recorded 44 and 35 SLG fault events in non-grounding and arc suppression coil grounding types, respectively. These faults were primarily caused by tree contact and fiber-optic cables bridging conductors. A total of 578 samples were collected from the real-time operational data of the two grounding configurations, including 274 fault samples. Among them, 163 samples were obtained from the non-grounding type and 111 from the arc suppression coil grounding type. The experimental results show that the MDNP system achieved an average fault line selection accuracy of 97.3% and

Table 11
Results of the fault line selection online application of the MDNP system.

Neutral grounding types	Number of distributions	Retraining time (s)	Detecting time (s)	Detection Accuracy
				MDNP (Ours)
Non-grounding	94	200~240	1.78	0.982
Arc suppression coil	29			0.973

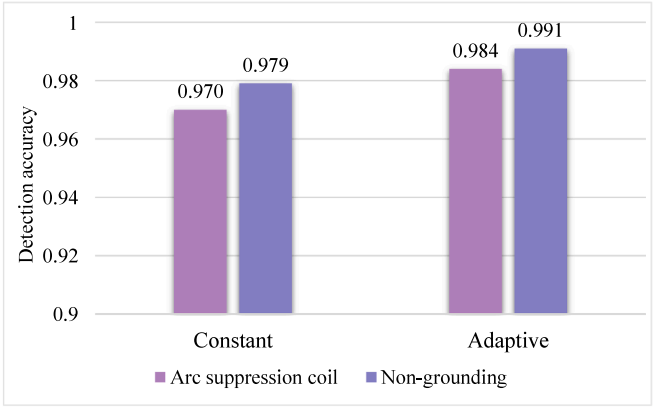


Fig. 15. Comparison of fault line selection accuracy of the MDNP system with constant and adaptive learning rates under different grounding types.

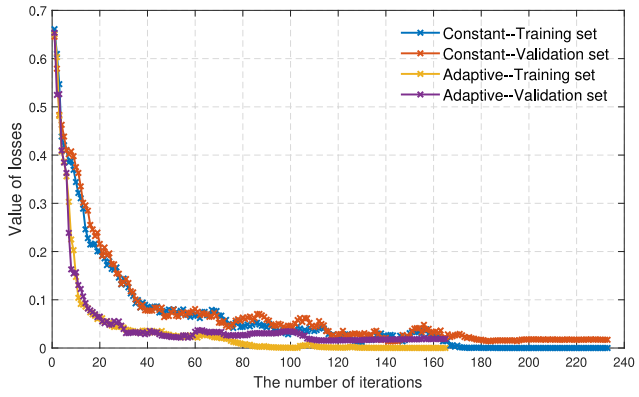


Fig. 16. Comparison of the iterations of the MDNP System with constant learning rate and adaptive learning rate.

98.2% under the two grounding types, respectively. After a period of operation, the average time for model retraining is approximately 200~240s. Additionally, the average response time from data collection to result output was 1.78s, verifying the effectiveness and real-time performance of the method in practical engineering applications.

For the purpose of verifying the detection performance of the proposed MDNP system under different network conditions, this paper selected two 35 kV substations with differing feeder configurations within a specific area as field measurement case studies. Substation I operates with a non-grounding neutral type and comprises seven 10 kV outgoing lines (specifically Line1-Qingchong, Line2-Qinggong I, Line3-Qinggong II, Line4-Qingma, Line5-Qingguan, Line6-Qingxin, and Line7-Qingyang); an SLG fault occurred at this station at 13:00:40 on March 15, 2022, caused by tree felling. Substation II operates with a neutral grounding via an arc suppression coil and includes five 10 kV outgoing lines (specifically Line1-Shapo, Line2-Guijinnuo, Line3-Haitang, Line4-Shenping, and Line 5-Changle); an SLG fault occurred at this station at 14:10:54 on April 17, 2022, due to foreign object contact on the tower. Fig. 17 illustrates the fault line selection test results of the MDNP system in substations with varying numbers of outgoing lines. The MDNP model achieved detection accuracies of 0.987 and 0.971 for the faulty lines, Line4-Qingma and Line3-Haitang. For healthy lines, the output values remained consistently below 0.035. These results indicate that the MDNP system is robust to substation topology changes and has strong generalization capability.

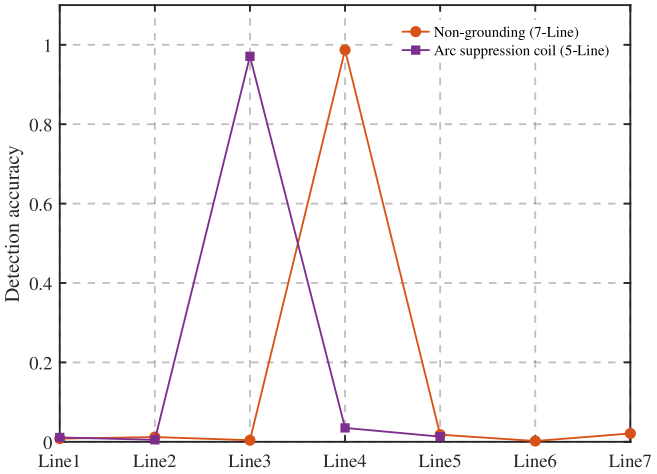


Fig. 17. The results of fault line selection tests for the MDNP system in substations with varying numbers of lines.

6. Conclusions

This paper proposes a novel MDNP system based on dendritic computation mechanisms, providing a solution for the fault line selection problem in distribution networks with multi-ground types. The MDNP system overcomes the limitations of traditional SNP systems, which use point neuron models, by fully utilizing the independent computational ability of biological neuronal dendritic branches. It constructs a dendritic hierarchical computation architecture with hierarchical information processing capability. This architecture uses memory dendritic factor triggering rules and a dynamically adjustable multi-branch dendritic structure. It enhances feature learning from power scheduling data and improves performance across different grounding types. In addition, the grounding type is obtained based on the CIM model and a real simulation model is established to achieve data augmentation. During the training process of the MDNP system, an adaptive learning mechanism is used to improve the system's convergence speed and detection accuracy. In terms of practical application, an online application scheme is developed, which dynamically optimizes the MDNP system using real-time data. The system's engineering practicality has been validated through deployment in the Guangxi power grid, demonstrating that it can effectively assist power personnel in accurately identifying SLG faulted lines.

For future work, the power scheduling data employed herein is a steady-state effective value with an update frequency of 2–4 s. Consequently, the background noise considered in this study is simulated as long-term stable Gaussian white noise in the system environment. To ad-

vance the research, this paper will additionally incorporate the influence of different background noise types to evaluate model performance.

CRediT authorship contribution statement

Song Ma: Writing – original draft, Validation, Methodology; **Fei Li:** Writing – review & editing, Project administration, Methodology; **Xiu Yin:** Methodology, Data curation, Conceptualization; **Xiyu Liu:** Writing – review & editing, Supervision; **Yangheng Hu:** Software, Investigation, Data curation; **Xing Zhang:** Writing – review & editing, Supervision, Project administration; **Mingyao Ma:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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